PURDUE UNIVERSITY

CE 29800 – BASIC MECHANICS II (DYNAMICS)

Course Information

Course number and title	CE 29800-002 Basic Mechanics II (Dynamics)
CRN	19150
Instructional Modality	Face-to-Face
Course credit hours	3
Prerequisites	CE 297 Basic Mechanics I (Statics) or equivalent (min grade C-) MA 261 Multivariate Calculus (min grade C-)

Course Description

Catalog Description: Kinematics of particles. Kinetics of particles and systems of particles. Kinematics of rigid bodies. Mass moments of inertia. Kinetics of rigid bodies. Mechanical vibrations.
Objective: to understand and learn to express dynamic motion of particles, collection of particles, and bodies; forces and energy involved.

Learning Outcomes

With successful completion of this course the student shall be able to:

- Relate displacement, velocity, acceleration, and time without reference to the cause of the motion for particles in 3-dimensional motion and rigid bodies in plane motion.
- Relate the 3-dimensional motion of particles and plane motion of rigid bodies to the applied forces.

Course addresses ABET Student Outcome: 1
1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Learning Resources, Technology & Texts

Textbook (Required)
Beer, F.P., Johnston, E.R., Jr., Cornwell, P.J., and Self, B. Vector Mechanics for Engineers: Dynamics, 12th Edition, McGraw-Hill, Boston, 2018. ISBN: 978-1259977305
(11th, 10th, 9th, 8th, 7th and 6th editions of this text are also acceptable. International SI only versions are not acceptable because they do not have US customary units. Hardcopy, e-textbook, pdf versions are all acceptable.)

Brightspace
Brightspace will be the main form of out-of-class interaction for this course. Students are encouraged to familiarize themselves with the process of viewing and submitting quizzes and obtaining handouts through Brightspace.

Course Outline [BJCS, 12th edition]

1. Introduction
2. Kinematics of Particles: (Chapter 11)
    - Rectilinear motion (11.1)
    - Special Cases and Relative Motion (11.2)
    - Curvilinear motion (Rectangular components) (11.4)
    - Tangential and normal components (11.5A)
    - Radial and transverse components (11.5B)
3. Kinetics of Particles: Newton's Second Law (Chapter 12)
    - Equations of motion (12.1)
    - Angular momentum, central forces (12.2)

4. Kinetics of Particles: Energy and Momentum Methods (Chapter 13)
    - Work, potential energy, kinetic energy, and power (13.1)
    - Conservation of energy, central forces (13.2)
    - Impulse and momentum, impulsive motion (13.3)
    - Impact, mixed problems (13.4)
5. *Systems of Particles (Chapter 14)*
    - *Motion of systems of particles (14.1)*
    - *Energy and momentum methods for systems of particles (14.2)*
6. Kinematics of Rigid Bodies (Chapter 15)
    - Translation, rotation about a Fixed Axis (15.1)
    - Velocities in plane motion (15.2)
    - Instantaneous center (15.3)
    - Accelerations in plane motion (15.4)
    - Coriolis acceleration in plane motion (15.5)
7. Moments of Inertia of Masses (Appendix B.1-5)
8. Plane Motion of Rigid Bodies: Forces and Accelerations (Chapter 16)
    - Plane motion of rigid bodies (16.1)
    - Constrained plane motion (16.2)
9. Plane Motion of Rigid Bodies: Energy and Momentum Methods (Chapter 17)
    - Work and energy (17.1)
    - Impulse and momentum, conservation of angular momentum (17.2)
    - Impulsive motion, eccentric impact (17.3)
10. Mechanical Vibrations (Chapter 19)
    - Vibrations of particles (19.1)
    - Vibrations of rigid bodies (19.2)
    - Energy methods (19.3)
    - Forced vibrations (19.4)

Note: Numbers in parenthesis given above are the chapter and section numbers from the textbook (12th edition).

Course outline may be revised during the semester.

Assignments and Weights

Assignments	Date	Weight
Quizzes	weekly	25%
Term Exam #1	*TBA*	25%
Term Exam #2	*TBA*	25%
Final Exam	*TBA*	25%

- Quizzes will be assigned and submitted via Brightspace. Two lowest quiz scores will be dropped from course quiz performance score calculation.
- Term exams and Final exam will be administered as in-person evening exams. You may be asked to show your Purdue ID Card in exams. Taking all exams is necessary (required) but not sufficient to receive a passing grade.
- Quiz/exam grade correction requests must be submitted in writing within one week after the grade is posted.

The instructor reserves the right to hold a meeting with a student to verify the authenticity of level of understanding as implied by solution(s) provided by the student in an assessment (quiz, exam) before finalizing a grade for that assessment. Note that this authenticity verification may take place after posting of an initial grade posting.

Should you miss an assignment, you will need to submit a doctor's report, or the instructor be informed formally and directly by the University Administration about your legitimate excuse, to be allowed to have a make-up assignment. In such cases, an arrangement for make-up will be made between the student and the instructor.

Homeworks
No homework will be assigned.

Practice Problems
Practice problems will be posted so that you may try applying your understanding of course material. You may work with your classmates, course tutors, course TA and the course instructor on practice problems.

Collaboration Policy
Quizzes: You may inquire about the quiz problems with the course TA and the course instructor only. You may not discuss the quiz problems with anyone else (i.e., no discussion with your classmates or course tutors). **Copying solutions from others in class or from an outside source (human, so-called "artificial intelligence", websites, or otherwise) is an act of cheating and a violation of the Purdue Academic Integrity and course honor codes.** Zero tolerance for violators.

Term Exams and Final Exam: No collaboration, sharing, or cheating of any kind is allowed. **Copying solutions from others in class or from an outside source (human, so-called "artificial intelligence", websites, or otherwise) is an act of cheating and a violation of the Purdue Academic Integrity and course honor codes.** Zero tolerance for violators.

Grade Calculation
Your course score will be the largest of the following:

- 25% Course Quiz Score + 25% of Term Exam 1 Score + 25% of Term Exam 2 Score + 25% of Final Exam Score
- 25% Course Quiz Score + 25% of Term Exam 2 Score + 50% of Final Exam Score
- 25% Course Quiz Score + 25% of Term Exam 1 Score + 50% of Final Exam Score
- 25% Course Quiz Score + 75% of Final Exam Score

Grading Scale
Final grades will be based on standard grade division, at the discretion of the instructor:

$$A \geq 90 \mid 90 > B \geq 80 \mid 80 > C \geq 70 \mid 70 > D \geq 60 \mid 60 > F$$

Please note that when it comes to grades, what counts is not how much you think you know, or what you think you could do, could have done, should have done, would have done if certain things were different or how much you have progressed. What matters is what you demonstrate on paper in quizzes, term exams and final exam. Be aware of the consequences of failing in this course.

Academic Integrity
Purdue's Honor Pledge: *"**As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue.**"*

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by emailing Office of Student Rights & Responsibilities, calling 765-494-8778 or using the reporting link on https://www.purdue.edu/odos/osrr/academic-integrity/index.html. The latter two options allow anonymous reporting.

By registering for this course you agree to abide by the course honor code that **you shall never take unfair advantage of others in this course**. Purdue University academic policies and procedures apply:

Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty." [Part 5, Section III-B-2-a, University Regulations] Furthermore, the University Senate has stipulated that "the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest." [University Senate Document 72-18, December 15, 1972]

Incidents of academic misconduct in this course will be addressed by the course instructor and referred to the Office of Student Rights and Responsibilities (OSRR) for review at the university level. **Any violation of course policies as it relates to academic integrity will result in a failing grade (F) for the course.** In addition, all incidents of academic misconduct will be forwarded to OSRR, where university penalties, including removal from the university, may be considered.

Those who are caught cheating and those who aid and abet cheating directly or indirectly will receive F grade (Failing) as the course grade for academic dishonesty. This policy applies to transgressions in quizzes and exams.

No material from this course shall be shared with anyone outside this class in any form whatsoever (printed, digital, direct mailing, website posting, etc.). These material include handouts, lecture notes, lecture videos and other recordings, quiz problems and solutions, exam problems and solutions, practice problems and solutions, and the notes you take during the class. All of these material, including your notes are subject to the instructor's copyright. Note that your notes are considered to be 'derivative works' of the instructor's presentations and materials, and they are thus subject to the instructor's copyright in such presentations and materials. Commercial note taking is prohibited. Those who violate the "no sharing" policy will be reported to authorities and pursued to the full extent of the University regulations and the Law.

Attendance Policy

This course follows the University Academic Regulations regarding class attendance, which state that students are expected to be present for every meeting of the classes in which they are enrolled. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, you should inform the instructor of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to is not possible, contact the instructor as soon as possible by email. For absences that do not fall under excused absence regulations (see below), this course follows the following procedures:

1. Do not come to class if you are feeling ill, but send an email to the instructors at ayhan@purdue.edu with the **subject line: CE298 absence**. No need for details about your symptoms. Just let the instructor know you are feeling ill and cannot come to class. If it is an emergency situation, please follow the University regulations on emergent medical care (see below).
2. Unless it falls under the University excused absence regulations (see below), all quizzes should be submitted on time via Brightspace.
3. The most important consideration in any absence is how it will affect your achievement of the assignment objectives and the course learning outcomes.

For cases that fall under **excused absence regulations**, you or your representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for notification of the instructors. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or emergent medical care. The processes are detailed, so plan ahead.

Nondiscrimination Statement

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each and all to reach their potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies and Statements.

Accessibility

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let us, the instructors, know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: **drc@purdue.edu** or by phone at 765-494-1247.

Students with disabilities whose DRC Course Accessibility Letter (CAL) includes test accommodations must first release their CAL to us, the instructors, and then schedule to take their exams with the Purdue Testing Services at **https://www.purdue.edu/studentsuccess/testing-services/accommodated-testing/student.php**. You must schedule to take your exam at least one week before the exam date listed on the syllabus. In the case of exams during finals week, you must schedule to take your exam at least four weeks before the start of finals week. The instructors will provide the Purdue Testing Services with your exam and they will administer it and provide the result to the instructors for grade reporting. Students must inform the instructors immediately of cases where the Purdue Testing Services does not have space so that I can make other arrangements. Students who fail to follow this process and these deadlines risk not being able to have their accommodations for that exam.

Mental Health Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack**.** Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Basic Needs Security

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed. Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday.

Important Dates through the Semester

- Purdue's 2025-2026 academic calendar is available at
  https://catalog.purdue.edu/preview_program.php?catoid=18&poid=33739
- Purdue's add/drop calendar for 2025 Fall semester is available at
  https://catalog.purdue.edu/preview_program.php?catoid=18&poid=33633

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructors' control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructor or the TA via email. You are expected to read your @purdue.edu email on a frequent basis.

- For any emergency, text or call 911.
- There are more than 300 Emergency Telephones (aka blue lights) throughout campus that connect directly to the Purdue Police Department (PUPD). If you feel threatened or need help, push the button and you will be connected right away.
- If we hear a fire alarm, we will immediately suspend class and evacuate the building, proceed outdoors, away from the building. Do not use the elevator.
- If we are notified of a Shelter in Place requirement for a tornado warning we will suspend class and shelter in the lowest level of this building away from windows and doors.
- If we are notified of a Shelter in Place requirement for a hazardous materials release, we will shelter in our classroom shutting any open doors and windows.
- If we are notified of a Shelter in Place requirement for an active threat such as a shooting, we will shelter in a room that is securable preferably without windows.